Using deep learning to generate synthetic tax records

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What do we want?

Augmented IRS microdata for historical and future years

What do we have?

- **One year of tax data, filers only**
  - About ten years out of date
  - Some values are blurred and some records are distorted
  - No non-filers, and none of the other variables a tax calculation needs: consumption, wealth, unrealized gains, estates
- **Surveys** that hold some of this information, but noisily and not at administrative quality
- **Aggregate targets**
  - Past years: IRS totals by AGI group, filing status and age range
  - Future years: CBO-style projections of 1040 items, or of total wages, business profits and similar totals

The task

Turn these scattered sources into

synthetic, realistic, augmented IRS microdata

for every year

How has this been done before?

	Status	Weight	Wages	Interest	Capital gains
Filers (PUF)	Married	812	$142k	$310	$5.2k
	Single	1,404	$38k	$0	$0
	Married	655	$0	$4.9k	$61k
	Head of hh.	1,120	$51k	$12	$0
	Single	3	$2.3m	$88k	$3.9m

1. **Start with the IRS PUF**, whatever year is available.

Made-up records.

How has this been done before?

	Status	Weight	Wages	Interest	Capital gains
Billionaires	Married	1	$1.2m	$14m	$310m
Filers (PUF)	Married	812	$142k	$310	$5.2k
	Single	1,404	$38k	$0	$0
	Married	655	$0	$4.9k	$61k
	Head of hh.	1,120	$51k	$12	$0
	Single	3	$2.3m	$88k	$3.9m
Non-filers	Single	2,310	$0	$0	$0
	Single	1,870	$6.5k	$0	$0

2. **Add records**: non-filers, and the very rich the PUF leaves out.

Made-up records.

How has this been done before?

	Status	Weight	Wages	Interest	Capital gains	Net worth	Consump-tion	Tips
Billionaires	Married	1	$1.2m	$14m	$310m	$4.1b	$6m	$0
Filers (PUF)	Married	812	$142k	$310	$5.2k	$610k	$96k	$0
	Single	1,404	$38k	$0	$0	$9k	$34k	$6.1k
	Married	655	$0	$4.9k	$61k	$2.4m	$88k	$0
	Head of hh.	1,120	$51k	$12	$0	$31k	$47k	$0
	Single	3	$2.3m	$88k	$3.9m	$38m	$410k	$0
Non-filers	Single	2,310	$0	$0	$0	$2k	$19k	$0
	Single	1,870	$6.5k	$0	$0	$1k	$15k	$1.2k

3. Add columns: estimate Y given X in a survey, where X is also on the PUF; draw Y for each record.

Made-up records.

How has this been done before?

	Status	Weight	Wages	Interest	Capital gains	Net worth	Consump-tion	Tips
Billionaires	Married	1	$1.2m	$14m	$310m	$4.1b	$6m	$0
Filers (PUF)	Married	860 ↑	$142k	$310	$5.2k	$610k	$96k	$0
	Single	1,290 ↓	$38k	$0	$0	$9k	$34k	$6.1k
	Married	702 ↑	$0	$4.9k	$61k	$2.4m	$88k	$0
	Head of hh.	1,180 ↑	$51k	$12	$0	$31k	$47k	$0
	Single	2 ↓	$2.3m	$88k	$3.9m	$38m	$410k	$0
Non-filers	Single	2,450 ↑	$0	$0	$0	$2k	$19k	$0
	Single	1,760 ↓	$6.5k	$0	$0	$1k	$15k	$1.2k

4. **Reweight to the target year**: grow weights for population and dollars for prices, then move each weight up or down until the year's totals are hit.

Made-up records.

What were the problems with this approach?

- **It did not work well for past years.** The affordability project made this clear.
- **A separate model for each variable**, each estimating its conditional relationships in its own way.
- **Reweighting worked for only a few past years.** Tax law changed what the IRS targets measure, mainly TCJA cutting back itemized deductions.
- **Stretched further every year.** Plausible records for future years, but worse with each year away from 2015.

The ideal approach

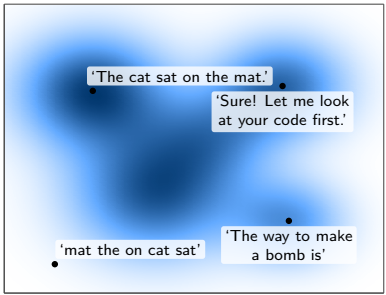
1. Take in microdata covering the joint relationships among every variable we might want to model.
2. Learn all of those relationships in **One Big Joint Probability Distribution**.
3. For any target year, supply (a) beliefs about which records correspond best to that year and (b) known statistical moments describing that year, and pull out of it the records that match them.

The result: a realistic synthetic microdata set for that year.

From a conversation with John and Harris.

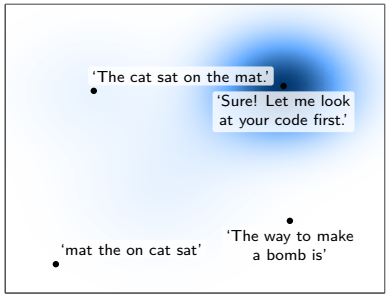
Language models: pretraining, then post-training

Pretraining



Read the whole internet and learn which sentences are likely: a prior over language

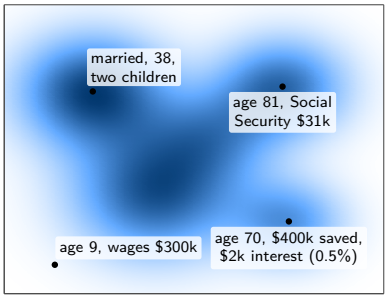
Post-training



Squeeze the mass toward a helpful, harmless assistant, and away from nonsense and harm

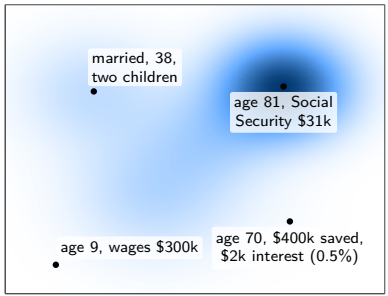
Our model: pretraining, then post-training

Pretraining



Learn which records are likely, from every source: a prior over records

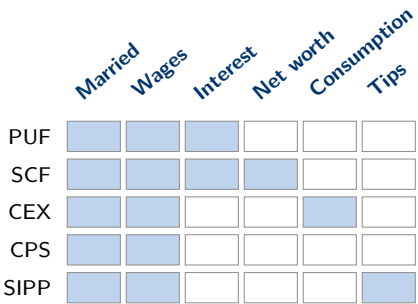
Post-training for 2040



Squeeze the mass toward records that hit 2040's totals, each still coherent

Post-training finds the distribution closest to the prior that hits the year's totals.

Pretraining



Shaded: the variable is in that source.

- As with a language model, all records from every source go into one training data set.
- One neural network is trained on the whole data set. It learns the distribution of each variable conditional on any set of other variables.
- A model that gives every such conditional distribution represents the full joint distribution, for example:

$$f(\text{married, wages, net worth}) = f(\text{married}) \times f(\text{wages} \mid \text{married}) \times f(\text{net worth} \mid \text{married, wages})$$

Model structure

- The model is a neural network, which is a set of connected logit-like functions arranged in layers. Each unit takes a weighted sum of its inputs and applies a simple nonlinear transformation. The next layer uses those outputs as its inputs.
- The final layer is a multinomial logit over the 84 ranges of the variable being predicted.
- The network is a transformer, the type used in language models. In a transformer, the prediction for each variable puts more weight on the other observed variables most relevant to it. Variables on the record play the role that words play in a language model.
- Our model has 4 layers and about 1.4 million parameters.

Model estimation

1. Draw a record from the training data. The source and year are always observed.
2. Hide a random subset of the record's variables.
3. Compute the model's predicted distribution for each hidden variable, conditional on the observed ones.
4. Compute the log probability assigned to each true value, multiplied by the record's survey weight.
5. Adjust each parameter by a small amount in the direction that increases this sum.

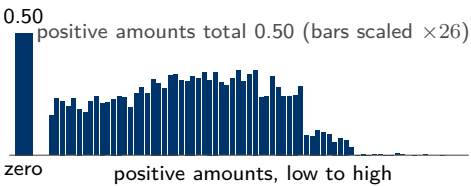
This is repeated over the full training data 80 times.

$$\max_{\theta} \sum_{i} w_i \sum_{j \text{ hidden}} \log p_\theta\left(x_{ij} \mid \text{shown}_i\right)$$

- The objective is a weighted sum of conditional log likelihoods, so the estimator is similar to maximum likelihood.
- The parameters are estimated by gradient descent.
- High-income records receive additional weight.
- Variables missing from a source are never predicted for that source's records.

Model output for one variable

Distribution of mortgage interest for married PUF filers in 2015
with AGI between the 75th and 90th percentiles



- Each dollar amount is divided into 84 ranges: zero, 10 ranges for losses and 73 ranges for positive amounts. The ranges are narrower at the top.
- The model gives a probability for each range rather than a single predicted value.
- The inputs are the data source, the year and any other variables known for the unit.

Probabilities are shares of records in the latest run's generated sample.

Sentence generation in a language model

Sentence so far

word 1 'The'

The sentence begins with 'The'.

The model then gives a probability for each possible next word.

Made-up probabilities.

Sentence generation in a language model

Sentence so far

word 1 'The'
word 2 'school'

Next word?



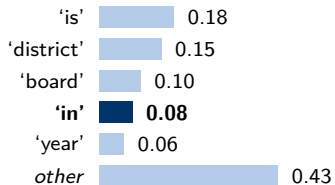
Made-up probabilities.

Sentence generation in a language model

Sentence so far

word 1 'The'
word 2 'school'
word 3 'in'

Next word?



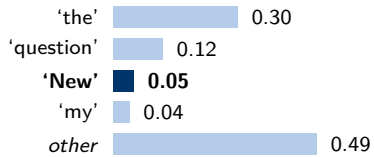
Made-up probabilities.

Sentence generation in a language model

Sentence so far

word 1	‘The’
word 2	‘school’
word 3	‘in’
word 4	‘New’

Next word?



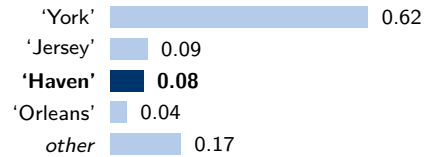
Made-up probabilities.

Sentence generation in a language model

Sentence so far

word 1	‘The’
word 2	‘school’
word 3	‘in’
word 4	‘New’
word 5	‘Haven’

Next word?



The drawn word is not always the most likely one.

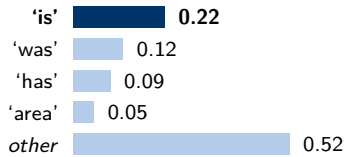
Made-up probabilities.

Sentence generation in a language model

Sentence so far

word 1	‘The’
word 2	‘school’
word 3	‘in’
word 4	‘New’
word 5	‘Haven’
word 6	‘is’

Next word?



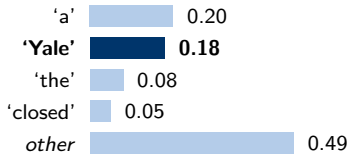
Made-up probabilities.

Sentence generation in a language model

Sentence so far

word 1	‘The’
word 2	‘school’
word 3	‘in’
word 4	‘New’
word 5	‘Haven’
word 6	‘is’
word 7	‘Yale’

Next word?



Each word is drawn conditional on all previous words. Records are generated in the same way.

Made-up probabilities.

Record generation

Record so far

source	PUF
year	2015
filer	yes
AGI group	75th–90th percentile

The source, year, filer status and AGI group are fixed before any draws.

The model then gives a probability for each value of the next variable.

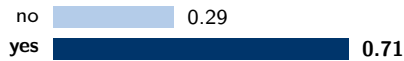
Probabilities are shares of records in the latest run's generated sample that match the record so far.

Record generation

Record so far

source	PUF
year	2015
filer	yes
AGI group	75th–90th percentile
married	yes

Next variable: married



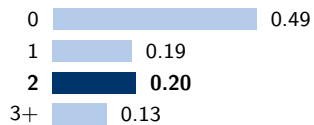
Probabilities are shares of records in the latest run's generated sample that match the record so far.

Record generation

Record so far

source	PUF
year	2015
filer	yes
AGI group	75th–90th percentile
married	yes
dependents	2

Next variable: dependents



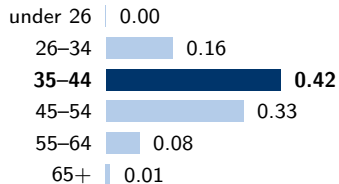
Probabilities are shares of records in the latest run's generated sample that match the record so far.

Record generation

Record so far

source	PUF
year	2015
filer	yes
AGI group	75th–90th percentile
married	yes
dependents	2
age range	35–44

Next variable: age range



Probabilities are shares of records in the latest run's generated sample that match the record so far.

Record generation

Record so far

source	PUF
year	2015
filer	yes
AGI group	75th–90th percentile
married	yes
dependents	2
age range	35–44
filing status	joint
child credit kids	2

Filing status and the number of child tax credit children are drawn in the same way.

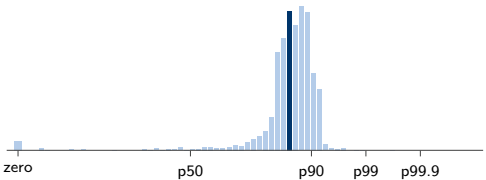
Probabilities are shares of records in the latest run's generated sample that match the record so far.

Record generation

Record so far

source	PUF
year	2015
filer	yes
AGI group	75th–90th percentile
married	yes
dependents	2
age range	35–44
filing status	joint
child credit kids	2
wages	82nd–84th pct. band

Next variable: wages



The drawn value (dark) is the band from the 82nd to the 84th percentile of wage earners.

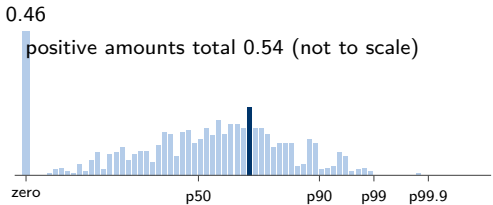
Probabilities are shares of records in the latest run's generated sample that match the record so far.

Record generation

Record so far

source	PUF
year	2015
filer	yes
AGI group	75th–90th percentile
married	yes
dependents	2
age range	35–44
filing status	joint
child credit kids	2
wages	82nd–84th pct. band
mortgage int.	66th–68th pct. band

Next variable: mortgage interest



The drawn value (dark) is the band from the 66th to the 68th percentile of filers with mortgage interest.

Probabilities are shares of records in the latest run's generated sample that match the record so far.

Record generation

Record so far

source	PUF
year	2015
filer	yes
AGI group	75th–90th percentile
married	yes
dependents	2
age range	35–44
filing status	joint
child credit kids	2
wages	82nd–84th pct. band
mortgage int.	66th–68th pct. band
dividends	40th–42nd pct. band
Schedule C	zero
···	about 90 tax items

The remaining tax variables are drawn in the same way, in a random order. Totals such as AGI are computed as sums of the drawn components.

Survey variables are drawn next: the balance sheet from the SCF and consumption from the CEX.

Each range is converted to dollars by drawing a real amount from that range and growing it to the target year.

Probabilities are shares of records in the latest run's generated sample that match the record so far.

The generated sample

- 400,000 filer records are drawn as the PUF would record them, with a fixed number in each of 12 AGI groups.
- Each record's starting weight is the number of returns in its AGI group divided by the number of records drawn in that group.
- 54,379 non-filer records are drawn as the CPS would record them.
- Each record also receives survey variables: the balance sheet from the SCF, consumption from the CEX and tips from the SIPP.
- Each record is a new draw from the joint distribution rather than a copy of a survey household.

Latest run, 2019 file.

Post-training: reweighting to targets

Records are kept as drawn. New weights w_i are chosen to stay as close as possible to the starting weights w_i^0:

$$\min_w \sum_i w_i \log \frac{w_i}{w_i^0} \quad \text{such that every target is within } \pm 1 \text{ standard error (SCF: } \tfrac{1}{2}\text{)}$$

- This is the same distance measure used in raking.
- For 2019 there are 753 targets, all of which are met. They include IRS tables by AGI, filing status and age, business income, W-2 tips, non-filer counts, SCF wealth and CEX spending.

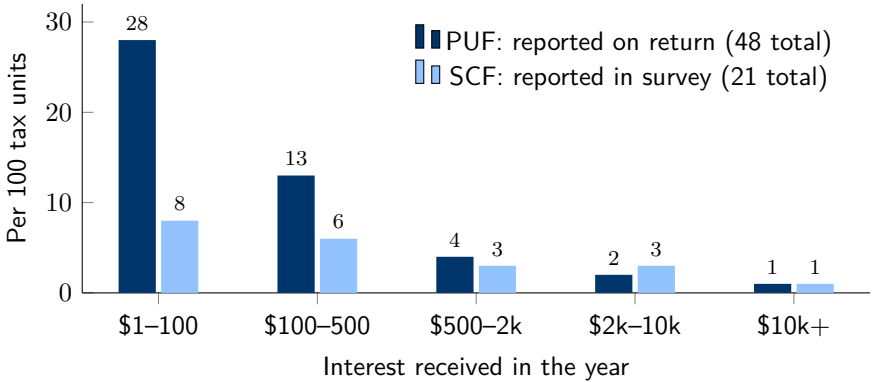
Distance of weight movement

Ratio of new to starting weight within a factor of	Share of records	Share of returns
1.05	15%	20%
1.1	26%	36%
1.25	44%	61%
1.5	58%	77%
2	70%	86%
4	81%	92%
10	96%	99%

- A factor of 2 means the new weight is between half and double the starting weight.
- The effective sample size is $(\sum w)^2 / \sum w^2$. It equals the number of records when all weights are equal and declines as the weights become more unequal.
- Reweighting retains 87 percent of the effective sample size, which falls from 280,800 to 244,700 (out of 454,379 records).

Latest run, 2019 file. Share of returns is measured with starting weights.

Challenge: measurement differences between surveys and tax returns



The gap is concentrated in small amounts, which often go unreported in the survey. Above $2,000 the two sources agree.

Married, wages $75–200k. 2015 PUF and 2016 SCF (tax year 2015), weighted.

The fix: assigning values for false negatives

SCF household	interest, as reported	interest, as on a return
household 1	$250	$250
household 2	$0	?
household 3	$0	?
household 4	$1,800	$1,800

- Each SCF household has two versions of interest, dividends and capital gains.
- The first version is the survey answer, which is never changed.
- The second version is the amount that would appear on a tax return. Households that report an amount keep it. Households that report zero may be assigned a value.
- PUF records have only the tax-return version, so both sources use the same definition.

Assigning values for false negatives during pretraining

Per 100 couples	PUF	SCF	Fill in
\$1–100	28	8	20
\$100–500	13	6	7
\$500–2k	4	3	2
\$2k and over	3	3	0
Any	48	21	28

The PUF determines the number and size of fill-ins. The rest of the SCF record determines the households that receive them.

1. At the start of each pass through the data (from the third), the model computes each zero-reporting household's probability of holding an amount in each range.
2. This probability is the model's probability that the household reports that range, given the rest of its record, times (PUF share ÷ SCF share − 1) for its type (married, dependents, wage band), divided by its probability of reporting zero.
3. Each batch of records draws new fill-ins from these probabilities. The model is trained on the tax-return version of the variable.
4. The probabilities use only the reported values and the PUF, never earlier fill-ins.

Table: married, wages \$75–200k; 2015 PUF and 2016 SCF, weighted. Probabilities are capped at 100 percent.

Challenge: sample weights

Problem

PUF record	Returns it stands for
AGI below the median	about 1,390
AGI at p99.99 and up	about 1.6

With survey weights, the model sees very little of the top, which is 0.02 percent of returns. With equal weights, the model learns the sample rather than the population.

Approach: grouped weighting

1. Records are divided into groups: 12 AGI groups in the PUF and 5 income classes in the SCF.
2. In estimation, each record is weighted by its survey weight times q, where q is the number of records divided by the total survey weight in its group.
3. Within a group, records are weighted by survey weight. Across groups, they are weighted by record count. The top AGI group rises from 0.02 to 4.6 percent of the PUF's estimation weight.
4. In generation, a fixed number of records is drawn in each group and weighted by $1/q$, which reverses the adjustment exactly.

Other challenges

- **Tax unit construction.** Surveys record households. Each household is divided into tax units using a common set of rules. A filing model then determines whether each unit files.
- **Changes in tax law.** TCJA reduced the number of itemizers. New items such as the tips and overtime deductions do not appear on any past return.
- **The top of the distribution.** Every source has few very high-income records.
- **Future years.** There is no microdata for future years, only a small number of projected totals.
- **Validation.** No source contains every variable, so some relationships cannot be checked against data.

Status

- A 2019 file with 400,000 filer records and 54,379 non-filer records
- Five sources: the 2015 PUF, SCF, CEX, CPS and SIPP. Each survey is divided into tax units using the same rules.
- All PUF variables, plus a balance sheet, consumption, tips and auto loan interest
- 753 targets, all met within their standard errors, including IRS tables by AGI, filing status and age, the top 1, 0.1 and 0.01 percent, W-2 tips and SCF wealth

Next steps

- Rule-based variables added after generation: credit eligibility, mortality probabilities, unrealized gains and capital gains basis
- Output files in Tax-Simulator's input format, followed by a first Tax-Simulator run
- A tax calculator within post-training, to allow targets on computed tax
- Other years: 2015 to 2023, followed by the projection years
- The top of the distribution: the PUF's aggregate records and the Forbes 400

Questions